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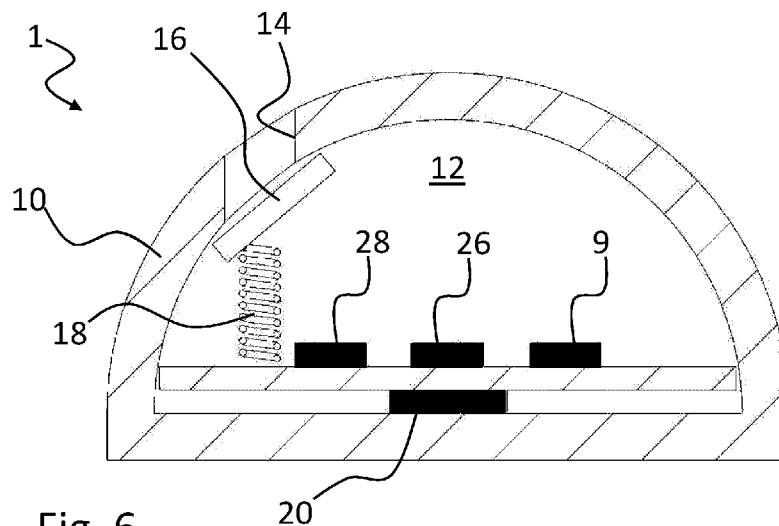


Fig. 6

(57) Abstract: The present invention relates to a device (1) for the recognition of a food and for the determination of the degree of freshness thereof, configured to be at least partially inserted inside a container (3) for foods, which can be closed by means of a closure (5), wherein said container (3) is adapted to contain a food (7). According to the invention, the device (1) comprises an external casing (10) within which at least one cavity (12) is defined, a wall of said external casing (10) having an opening (14) adapted to put the volume of said cavity (12) in fluid communication with the space outside said device (1), said opening (14) being hermetically closable by means of a closing element (16), said cavity (12) being provided with at least one gas detection device (9) configured to generate a signal indicative of the concentration of gases, said device (1) comprising a processing and control unit (20) in data communication with said gas detection device (9) configured to generate, on the basis of said signal indicative of the concentration of gases, a signal indicative of the type of food (7) present in said container (3) and a signal indicative of the degree of freshness of said food (7) present in said container (3), said device (1) being switchable between: - a setting configuration, in which said closing element (16) hermetically

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closes said opening (14) and in which said gas detection device (9) generates a signal indicative of the concentration of gases present exclusively inside said cavity (12), and - a detection configuration, in which said device (1) is at least partially inserted inside said container (3) so that said opening (14) faces the internal volume of said container (3) and in which said opening (14) is not occluded by said closing element (16) so that said cavity (12) is in fluid communication with the internal volume of said container (3), and in which said gas detection device (9) generates a signal indicative of the concentration of gases present inside said cavity (12) and inside said container (3).

**DEVICE FOR THE RECOGNITION OF A FOOD AND FOR THE DETERMINATION
OF THE DEGREE OF FRESHNESS THEREOF**

FIELD OF THE INVENTION

The present invention relates to a device for the recognition of a food and for the determination of the degree of freshness thereof. In particular, the present invention relates to a device configured to recognise the type of food present in a container and to monitor the degree of freshness or rottenness thereof.

STATE OF THE ART

As is well known, the need to minimise food waste, for environmental, social and economic reasons, is currently greatly felt both at the household level and in the catering sector.

Presently, food is preserved in refrigerators, often inside closable bags or containers, so as to prolong the preservation thereof in a state of freshness. However, it is equally well known that different types of food lose freshness at significantly different times from each other. Furthermore, even small variations in the preservation conditions often result in food rotting at times which are not entirely predictable and often do not correspond with the indications given at the time of purchase of the product with regard to the expiry date, or with the preferred date of consumption from the opening of the package.

Presently there are smart fridges, possibly connected to a mobile phone or other electronic device, which can keep track of the food they contain and the expiry date thereof and which, when equipped with internal cameras, can show to the owner, by means of a mobile phone, the contents thereof. The loading of useful information related to the specific food, in the memory unit of the refrigerator, i.e., the loading of information related to the type of food inserted and the expiry date thereof, requires an operation on the part of the user and/or the refrigerator, which may for example consist of:

- scanning the barcode or QR code associated with the individual product, which can be scanned by means of a special interface with which the smart refrigerator can be provided, or by means of the mobile phone which can be connected to the refrigerator;
- in the recognition, by the refrigerator itself, of an RFID tag associated with the product;
- in the manual entry of the product description and the expiry date thereof, by means of the refrigerator interface or by means of a mobile phone, in particular in cases in which the product has neither RFID tags nor barcodes containing the necessary information.

However, such known systems for monitoring the expiry date of foods are not without drawbacks, including, as mentioned above, the fact that the degree of freshness of a food is not

always directly related to the expiry date provided by the manufacturer, and may therefore provide unreliable indications to the user as to the actual freshness of the food.

Another drawback of such known systems is that a manual intervention by the user is always required, often not intuitive or even immediate, which often results in the user not using the tool available to him or her efficiently, or only making partial use thereof (e.g., only for products which do not require the manual entry of the expiry date), or even, over time, abandoning the use thereof altogether.

Yet another drawback lies in the fact that foods are generally not provided with barcodes or QR codes, let alone RFID tags, which contain and make the information needed to monitor the expiry date easily accessible.

Also known are devices for recognising the type of food and its state, such as those described in US2021/055275A1 and US2020/333310A1, in which gas detectors are employed which use adsorbent materials that are specific for the different types of gas to be detected and which vary the physical properties thereof (e.g., volume, mass, conductivity) on the basis of the amount of the specific gas adsorbed.

These devices are also not without drawbacks, among which should be underlined the fact that they are particularly complex to make and in the proper maintenance thereof during use. In fact, the technology of gas detection by means of adsorbent materials is highly specific to the type of gas to be detected, and therefore requires, on the one hand, prior knowledge of the type of gas to be detected, and on the other hand, the preparation of a set of different gas detectors if there are several gases of different types to be detected.

The technology of gas detection by means of adsorbent materials also requires complex and delicate procedures to extract the adsorbed gases from the adsorbent material so as to detect the presence and concentration thereof.

Furthermore, this technology is particularly sensitive to saturation problems of the adsorbent material by the gases to be detected and therefore requires constant and continuous monitoring of the performance of the detector itself, as well as the gas saturation level of the environment within which the detector operates.

SUMMARY OF THE INVENTION

In view of the above, the task of the present invention is to make a device for the recognition of a food and for the determination of the degree of freshness thereof which overcomes the limits of prior art, allowing the automatic recognition of the type of food and determination of the actual degree of freshness thereof.

Within the scope of this task, it is the object of the present invention to make a device

which is simple and intuitive to use, and requires minimal intervention by the user.

Another object of the present invention is to make a device which is capable of determining the degree of freshness of a food with extreme accuracy and repeatability.

5 A further object of the present invention consists in making a device which operates effectively with all different types of food.

A further object of the invention consists in making a device which is capable of giving the broadest guarantees of reliability and security when used.

Yet another object of the invention consists in making a device which is easy to manufacture and economically competitive.

10 The above-mentioned task, as well as the aforementioned objects and others which will become more clear below, are achieved by a device for the recognition of a food and for the determination of the degree of freshness thereof according to claim 1, as well as by a food preservation system according to claim 12 and a process for the recognition of a food and for the determination of the degree of freshness thereof according to claim 13.

15 Other features are provided in the dependent claims.

LIST OF FIGURES

Further features and advantages will become more apparent from the exemplary but non-limiting description of a preferred embodiment of the present invention, illustrated with the aid of the attached drawings in which:

- 20 - figure 1 schematically illustrates the device for the recognition of a food and for the determination of the degree of freshness thereof, according to the invention, inserted in a container containing a food;
- figure 2 is a perspective view of the device of figure 1, according to the invention;
- 25 - figure 3 schematically illustrates a first variant of the device of figure 1, according to the invention;
- figure 4 shows a second variant of the device of figure 1, according to the invention;
- 30 - figure 5 is a perspective view, sectioned with respect to a horizontal plane, of the device of figure 2, according to the invention;
- figure 6 is a vertical sectional view of the device of figure 2 according to the invention;
- figure 7 is a perspective view of a third variant of the device of figure 1, according to the invention, applicable to a food container.

DETAILED DESCRIPTION OF THE INVENTION

With particular reference to the figures, the device for the recognition of a food and for the determination of the degree of freshness thereof is indicated globally with the reference number 1.

5 According to the invention, such a device 1 is configured to be at least partially inserted inside a food container 3. The container 3 can be closed by means of a closure 5, such as a lid 50, or a cap 51, and is adapted to contain a solid or liquid food 7.

The device 1 comprises an external casing 10 within which at least one cavity 12 is defined. One wall of the external casing 10 has a through opening 14, adapted to put the volume
10 of the cavity 12 in communication with the space outside the device 1.

However, such an opening 14 can be hermetically closed by means of a closing element 16.

At least one gas detection device 9 configured to generate a signal indicative of the gas concentration is housed in the cavity 12.

15 The device 1 comprises a processing and control unit 20 in data communication with the gas detection device 9 configured to generate, on the basis of the signal indicative of the gas concentration, a signal indicative of the type of food 7 present in the container 3 and a signal indicative of the degree of freshness of the food 7 present in the container 3.

The device 1 can be switched between:

- 20 - a setting configuration, in which the closing element 16 hermetically closes the opening 14 and in which the gas detection device 9 generates a signal indicative of the gas concentration present exclusively inside the cavity 12 of the device 1;
- a detection configuration, in which (i) the device 1 is at least partially inserted inside the container 3 such that the opening 14 of the cavity 12 faces the internal volume of the
25 container 3, (ii) the cavity 12 is in fluid communication with the internal volume of the container 3, and (iii) the gas detection device 9 generates a signal indicative of the gas concentration present inside the cavity 12 and inside the container 3.

In essence, therefore, the device 1 has a setting configuration, or calibration configuration, in which the gas detection device 9 is in contact only with the gases present in
30 the cavity 12 of the device 1 itself, and a detection configuration, in which the opening 14 is open, and the cavity 12 is invaded by the gases also present outside the device 1, and in particular is invaded by gases emitted by the food 7 present inside the container 3 where the device 1 itself is also at least partially inserted.

As mentioned, the opening 14 is adapted to put the volume of the cavity 12 in fluid

communication with the space outside the device 1. In the use of the device 1, the opening 14 is positioned inside the container 3 so as to facilitate the entry of gases and air, with a degree of humidity thereof, inside the cavity 12, while preventing the entry of any liquids present into the container 3 which could possibly damage the internal components of the device 1.

5 Alternatively, the opening 14 could also be provided with a membrane permeable to gases but not to liquids.

Preferably, as also explained below, the device 1 is configured to remain in the setting configuration for a predetermined period of time, preferably comprised between 30 seconds and 1 hour, and even more preferably comprised between 1 minute and 20 minutes.

10 Preferably, in order for the device 1 to operate effectively, when the device 1 is operating in the detection configuration, the container 3 is closed by means of the closure 5. Preferably such a closure 5 hermetically closes the container 3. This thereby eliminates or at least minimises the infiltration of gases from the environment outside the container 3.

15 Preferably, the device 1 can be fully inserted inside the container 3. For example, as illustrated in figure 1, the device 1 is a stand-alone device which can be inserted into a container 3 as needed. As illustrated in figure 2, the device 1 can, for example, take the form of a hemisphere which can be inserted in the container 3. In this case, the opening 14 necessarily faces the internal volume of the container 3 where the food 7 to be monitored is present.

20 Alternatively, the device 1 can be integrated in the closure 5 of the container 3, so that when container 3 is closed by the respective closure 5, the opening 14 of the device 1 itself faces the internal volume of the container 3.

For example, as illustrated in figure 3, the device 1 can be integrated in the lid 50 of a container 3, so that it has a first part which, when the lid 50 is applied to the container 3, remains inside the container 3 itself, and a second part which instead remains outside the container 3. A
25 window 14 is obtained on the part of the external casing 10 which remains inside the container 3.

As illustrated in figure 7, the device 1 can be, at the same time, a stand-alone device, insertable inside any container 3, and a device which can be associated with the closure 5 of the container 3, for example inside a special seat 53 obtained in the lid 50 of the container 3.
30 Preferably, such a seat 53 is provided with openings 54 through which the device 1 can place itself in fluid communication with the internal volume of the container 3.

Preferably, the device 1 can also comprise an indicator light 84 configured to emit a light signal dependent on the signal indicative of the degree of freshness of the food 7. Advantageously, the indicator light 84 is in data communication with the processing and control

unit 20, which is configured to vary the light intensity and/or the colour of the light signal emitted by the indicator light 84 on the basis of the signal indicative of the degree of freshness of the food 7.

As illustrated in figure 4, the device 1 can also be integrated into a cap 51 of a liquid container, such as a bottle or carton of milk or fruit juice. The positioning of the device 1 in the cap 51 of a container 3 of liquids, therefore in the top position, prevents the contact of the liquid 7 contained in the container 3 with the device 1 itself, at least as long as the container 3 of liquids is in a vertical position.

The container 3 can therefore be a plastic container, transparent or opaque, made of food cardboard, or a bottle or even a bag for foods. The closure 5 can therefore, respectively, consist of a lid 50, a cap 51, or even a knot or a string applied to the end of a bag for foods.

The casing 10 can consist of one or more elements assembled together and can be made of different materials, such as plastic or metal.

Preferably, the device 1 comprises actuation means 18 configured to move the closing element 16 between an open position and a hermetically closed position of the opening 14, and vice versa.

As illustrated very schematically in figure 5, the closing element 16 can be defined by a shutter element of the opening 14 and the actuation means 18 can be defined by an elastic spring.

When switching from the setting configuration to the detection configuration, and vice versa, the actuation means 18 are activated or deactivated to open or close the opening 14.

Such actuation means 18 can be manually activated/deactivated by the operator in combination with mechanical or electronic timing mechanisms, or automatically on the basis of an activation/deactivation signal generated by the processing and control unit 20.

The opening 14 can basically change state, i.e., be closed or open, depending on the phase of the freshness monitoring procedure of the food 7. Such an opening 14 has the purpose of isolating and limiting the volume of the device 1 within which the gas concentration detection occurs during the setting, or calibration, step, so that when the opening 14 is hermetically closed, the calibration volume inside the device 1 itself and the volume of the container 3 outside the device 1 are isolated from each other. In the detection step, the opening 14 is instead open and the two volumes are in fluid communication with each other.

Preferably, the device 1 comprises a user interface, in data communication with the processing and control unit 20, configured to communicate to the user, on the basis of the signal indicative of the type of food 7 present in the container 3 and/or on the basis of the signal

indicative of the degree of freshness of the food 7 present in the container 3, the type of food 7 present in the container 3 and/or the degree of freshness of the food 7 present in the container 3.

Such a user interface can be integrated directly on device 1, where the processing and control unit 20 is also fully integrated.

The user interface may in particular be integrated on the device 1 in the embodiments of the invention in which the device 1 itself is integrated in the closure 5 of the container 3.

Preferably, the processing and control unit 20 is configured to transmit to an external electronic device 22, through a data communication interface 24, the signal indicative of the gas concentration, and/or the signal indicative of the type of food 7 present in the container 3 and/or the signal indicative of the degree of freshness of the food 7 present in the container 3.

The external electronic device 22 can be a mobile phone, a computer, a tablet, or a data server.

Preferably the user interface can reside on the external electronic device 22, and in particular it can reside on a mobile phone or tablet application, on a computer programme, or even reside on a web-app.

In this case the processing and control unit 20 can be split between an operational part on the device 1 itself and an operational part on the external electronic device 22. For example, the operational part on the device 1 can be configured to process the signal indicative of the concentration of gases and send it to the operational part on the external electronic device 22, where the processing and generation of the signal indicative of the type of food 7 present in the container 3 and/or the signal indicative of the degree of freshness thereof occurs.

Preferably, the gas detection device 9 comprises a volatile organic compound sensor, preferably of the silicon type with a highly sensitive metal oxide (MOx) chemiresistor.

Alternatively, the gas detection device 9 comprises a photo ionisation detector (PID) or a flame ionisation detector (FID).

As is well known, volatile organic compounds are those organic compounds which exhibit high volatility. Italian legislation defines 'volatile organic compounds' as those organic compounds which have a vapour pressure of 0.01 kPa or more at a temperature of 20°C.

The gas detection device 9 thus comprises a sensor capable of detecting the aforesaid volatile organic compounds generated by the food 7, and therefore of generating a signal indicative of the gas concentration of gases present in the container 3, i.e., the concentration of such aforesaid volatile organic compounds.

Preferably, the device 1 comprises a temperature sensor 26 configured to generate a

signal indicative of the temperature of the gases present inside the cavity 12. The processing and control unit 20 is in data communication with such a temperature sensor 26 and is configured to generate, on the basis of the signal indicative of the gas concentration and of the signal indicative of the temperature of the gases, a signal indicative of the type of food 7 present in the container 3 and a signal indicative of the degree of freshness of the food 7 present in the container 3.

In particular, the gas detection device 9 has an integrated circuit which, by means of the temperature control, detects the electrical conductivity of the metal oxides Mox to process the signal indicative of the gas concentration, which is a function of such electrical conductivity.

Preferably, the device 1 comprises a humidity sensor 28 configured to generate a signal indicative of the humidity of the gases present inside the cavity 12. The processing and control unit 20 is in data communication with such a humidity sensor 28 and is configured to generate, on the basis of the signal indicative of the gas concentration and of the signal indicative of the humidity of the gases, a signal indicative of the type of food 7 present in the container 3 and a signal indicative of the degree of freshness of the food 7 present in the container 3.

Thereby, the device 1 can use not only the gas concentration values, but also the gas temperature and/or humidity values to identify the type of food 7 present in the container 3 and to define the degree of freshness thereof.

At least a first classification algorithm is loaded on the processing and control unit 20 which is adapted to generate, at least on the basis of the signal indicative of the gas concentration, and preferably also on the basis of the signals indicative of the temperature and/or humidity of the gases, the aforesaid signal indicative of the type of food 7 present in the container 3.

At least a second classification algorithm is further loaded on the processing and control unit 20 which is adapted to generate, at least on the basis of the variation over time of the signal indicative of the gas concentration, and preferably also on the basis of the signals indicative of the temperature and/or humidity of the gases, the aforesaid signal indicative of the degree of freshness of the food 7 present in the container 3.

Preferably both the first and the second classification algorithm comprise artificial intelligence algorithms, preferably of the machine learning type.

Alternatively, or in addition, both the first and the second classification algorithm also process comparisons on the basis of databases and/or other external data sources, in which comparison data of gas emission of known and predefined food are stored and provided. In particular, also the databases on the basis of which the algorithms perform the classification to

generate the signals indicative of the type of food and the degree of freshness thereof, can be updated with machine-learning type algorithms.

Preferably, the signal indicative of the type of food 7 present in the container 3 comprises a signal indicative of the category, to which the food 7 belongs, identified among a plurality of predefined categories, and preferably also a signal indicative of the sub-category, to which the food 7 belongs, identified among a plurality of predefined sub-categories belonging to a respective category.

Preferably, the plurality of predefined categories comprises at least the following categories: meat, fish, fruit, vegetables, milk and milk derivatives, grains and grain derivatives.

By way of example, the sub-categories of the meat category could be: poultry, pork, beef, veal. The sub-categories of the fruit category could be related to different types of fruit, e.g., strawberries, kiwis, bananas. The sub-categories of the category milk and milk derivatives could distinguish between e.g., aged cheeses, fresh cheeses.

The signal indicating the type of food 7 present in container 3 could also comprise a signal related to further features of the food 7, such as the presence or absence of preservatives, or distinguish whether it is a raw or cooked food.

Preferably, the device 1 comprises a rechargeable battery or replaceable, disposable battery adapted to power all the electronic components present in the device 1 itself.

Preferably, so as to facilitate, on the part of the processing and control unit 20, the generation of the signal indicative of the type of food 7 present in the container 3 and also the signal indicative of the degree of freshness of the food 7, the device 1 comprises a selector 8 operable by the user to select the category to which the food 7 present in the container 3 belongs from a plurality of preset categories of belonging. The processing and control unit 20 is configured to receive in input an initial signal indicative of the category of belonging of the food 7 among the plurality of preset categories selected by means of the selector 8.

Preferably, the user's operation of the selector 8 to select the category of belonging of the food 7 also causes the device 1 to switch on.

Preferably, the selector 8 comprises an encoder associated with a rotatable wheel 80.

As illustrated in figure 7, the device 1 can comprise a selector 8 in the form of a rotatable wheel 80 which can be rotated and positioned in three different positions, so as to select a category to which the food 7 belongs among three preset categories, such as 'fruit, meat, rice,' indicated by special category symbols 81. A further fourth position of the selector 8, represented by the on/off symbol 82, can be selected to switch the device 1 on/off. In particular, when the indicator 83 of the wheel 80 is positioned at the on/off symbol 82, the device 1 is off. On the

other hand, when the indicator 83 of the wheel 80 is positioned at one of the category symbols 81, the device 1 is switched on and the processing and control unit 20 receives in input an initial signal indicative of the category of belonging of the food 7 among the preset categories.

5 In this case, the first and the second classification algorithm are both configured to generate the signals indicative of the type of food 7 and the degree of freshness thereof on the basis not only of the signal indicative of the gas concentration, but also on the basis of the initial signal indicative of the category of belonging of the food 7 present in the container 3 among a plurality of preset categories of belonging. Advantageously, therefore, the optional presence of the selector 8 makes it possible to simplify the first steps of data acquisition, thus increasing the
10 number of recognisable foods, as well as favouring an interaction with the user characterised by greater user involvement.

The present invention further relates to a food preservation system 100 comprising at least one device 1 for the recognition of a food and for the determination of the degree of freshness thereof as described above, and at least one respective container 3 for foods closable
15 by means of a closure 5.

The system 100 can advantageously comprise a plurality of devices 1 insertable in a respective plurality of containers 3 each containing a food 7. In this case, the plurality of devices 1 is connected to an external electronic device 22 which receives information related to each of the containers 3 equipped with the respective device 1, so as to allow the user to simultaneously
20 monitor the degree of freshness of a plurality of foods 7 placed in a respective plurality of containers 3, each containing a device 1 thereof.

As illustrated in figure 7, the container 3 can comprise a lid 50 in which a seat 53 is obtained and inside which the device 1 can be inserted. Such a seat 53 advantageously has, at the bottom, through openings 54 configured to face the opening 14 of the device 1, so as to put
25 the internal volume of the container 3 in fluid communication with the cavity 12 of the device 1 where the gas detection device 9 is placed.

The present invention further relates to a procedure for the recognition of a food and for the determination of the degree of freshness thereof, comprising the steps of:

- arranging a device 1 for the recognition of a food and for the determination of
30 the degree of freshness thereof as described above
- arranging a container 3 containing a food 7;
- setting the device 1 in the setting configuration for a predetermined period of time comprised between 30 seconds and 1 hour, preferably comprised between 1 minute and 20 minutes;

- inserting, at least partially, the device 1 inside the container 3 containing the food 7;
- setting the device 1 in the detection configuration;
- generating a signal indicative of the type of food 7 present in the container 3;
- 5 - generating a signal indicative of the degree of freshness of the food 7 present in the container 3.

In particular, the interaction between the device 1 and the user is designed so as to ensure an ease and immediacy of use, while at the same time optimising the battery life of the device 1 itself.

10 In particular, the monitoring of the food 7 by the user starts with the step of setting the device 1 in the setting configuration. Such a step comprises a user interaction with device 1 so as to close the opening 14. Such an interaction can, for example, comprise the clockwise rotation of a part of the device 1 with respect to a remaining part, and the consequent closing of the opening 14 by means of appropriate levers. In such a step, the detection of the
15 concentration of gases inside the cavity 12 is also started by means of the gas detection device 9, for a time comprised for example between 1 and 20 minutes. During this span of time, the gas detection device 9 is isolated from the external environment, and in particular from the internal volume of the container 3. This span of time allows the gas detection device 9 to achieve a measurement stability in a closed environment. In fact, the gas detection device 9 is preferably
20 of the chemiresistive type, i.e., it operates on the basis of the detection of an electrical resistance value (chemiresistance) which varies as a function of the presence of gases reacting with the resistance itself. The signal indicative of the concentration of gas is obtained starting from the analysis of such resistance values, and in particular from the analysis of the variation of such resistance values which occurs when the nature and concentration of gases with which the gas
25 detection device 9 comes into contact varies. Accordingly, the algorithms for generating the signals indicative of the type of food 7 and the degree of freshness thereof are also on the basis of the analysis of differential signals, indicative of variations occurring inside the cavity 12 and the container 3. For this reason, it is important that the gas detection device 9 is subjected to a calibration, which occurs on the basis of a stable gas volume, in a sufficient span of time for
30 the gas conditions to stabilise.

While the device 1 is in the setting configuration, it can be inserted inside the container 3 containing the food 7. The container 3 can then be closed by means of the closure 5. As soon as the setting time runs out, the device 1 is set to the detection configuration and the opening 14 opens. This can occur thanks to timing mechanisms, e.g., mechanical, activated by the initial

user interaction, or by electronically automated mechanisms. In the detection configuration, the gas detection device 9, and possibly the temperature 26 and humidity 28 sensors, measure the parameters of the gases present in the same environment in which the food 7 is also inserted.

On the basis of such information, the type of food 7 present in the container 3 is first
5 recognised, and the changes in the parameters of gas concentration, temperature and humidity are then recorded so as to process the signal indicative of the degree of freshness of the food 7.

When the user decides to stop the monitoring, he or she opens the closure 5 of the container 3 and interacts with the device 1 by closing the opening 14 again, so as to isolate the gas detection device 9 again from the external environment and make it ready for a new use. If
10 necessary, it is also possible to switch off the electronic components of the device 1 itself, to further reduce energy consumption.

The hermetic closure of the opening 14 of the device 1 at the end of use also allows the washing thereof without damaging the electronic components present therein.

In practice, it has been found that the device for the recognition of a food and for the
15 determination of the degree of freshness thereof, according to the present invention, fulfils the task as well as the intended objects in that it allows the automatic recognition of the food contained in the food container, or at least the category and/or sub-category to which the food belongs, as well as any other relevant features of the food, without requiring complex and expensive cameras or barcode, QR reading systems or RFID tag detection systems.

20 Another advantage of the device, according to the invention, is that the recognition of the type of food does not even require any data entry by the user.

A further advantage of the device, according to the invention, lies in the fact that it includes an extremely limited number of sensors, and above all in the fact that it includes sensors available at low cost and which do not require large amounts of data to be detected and
25 processed.

Another advantage of the device, according to the invention, is that it requires minimal energy consumption and that the sensors and electronics are miniaturised.

Still another advantage of the device, according to the invention, consists in the fact of being able to use differential signals, which are not associated with an absolute value of a
30 quantity, but with a variation of a quantity, thanks to the fact of having included a calibration volume, and a calibration time, within which and during which, the gas, temperature, humidity detection devices are isolated from the external environment and thus allow the respective signal acquisitions to stabilise the values thereof before starting the actual recognition step of the type of food and freshness monitoring step.

Yet another advantage of the device, according to the invention, is that it can be used with any type of food container, from box-like containers to bags to bottles.

5 A further advantage of the device, according to the invention, is that it can be used independently of the food preservation technique. The device can be used inside containers which can be placed in fridges or cold rooms, in pantries or warehouses, or still in any environment where foods are preserved.

Yet another advantage is that the device, according to the invention, allows for a 'retrofitting' of normal food preservation technologies, at a very low cost.

10 The device for the recognition of a food and for the determination of the degree of freshness thereof thus conceived is susceptible to numerous modifications and variations, all of which fall within the scope of the inventive concept.

Furthermore, all the details can be replaced by other technically equivalent elements.

In practice, any materials can be used according to requirements, as long as they are compatible with the specific use, the dimensions and the contingent shapes.

CLAIMS

1. Device (1) for the recognition of a food and for the determination of the degree of freshness thereof, configured to be at least partially inserted inside a food container (3), which can be closed by means of a closure (5), said container (3) being adapted to contain, in the internal volume thereof, a food (7), said device (1) comprising an external casing (10) inside which at least one cavity (12) is defined, a wall of said external casing (10) having an opening (14) adapted to put the volume of said cavity (12) in fluid communication with the space outside said device (1) and with said internal volume of said container (3), said opening (14) being hermetically closable by means of a closing element (16), at least one gas detection device (9) comprising a volatile organic compound sensor being housed in said cavity (12) and being configured to generate a signal indicative of the gas concentration, said device (1) comprising a processing and control unit (20) in data communication with said gas detection device (9) configured to generate, on the basis of said signal indicative of the gas concentration, a signal indicative of the type of food (7) present in said container (3) and a signal indicative of the degree of freshness of said food (7) present in said container (3), said device (1) being switchable between:

- a setting configuration, in which said closing element (16) hermetically closes said opening (14) and in which said gas detection device (9) generates a signal indicative of the gas concentration of gases present exclusively inside said cavity (12), and

- a detection configuration, in which said device (1) is at least partially inserted inside said container (3) so that said opening (14) faces said internal volume of said container (3) and in which said opening (14) is not occluded by said closing element (16) so that said cavity (12) is in fluid communication with said internal volume of said container (3), and in which said gas detection device (9) generates a signal indicative of the gas concentration of gases present inside said cavity (12) and inside said internal volume of said container (3),

said device (1) comprising:

- a first classification algorithm loaded onto said processing and control unit (20), said first classification algorithm being adapted to generate, on the basis at least of said signal indicative of the gas concentration, said signal indicative of the type of food (7) present in said container (3), and
- a second classification algorithm loaded onto said processing and control unit (20), said

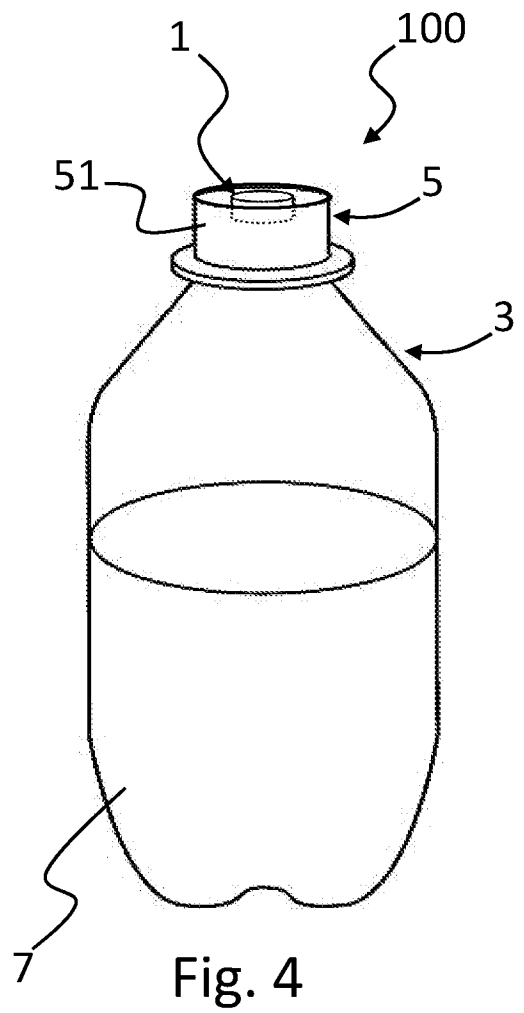
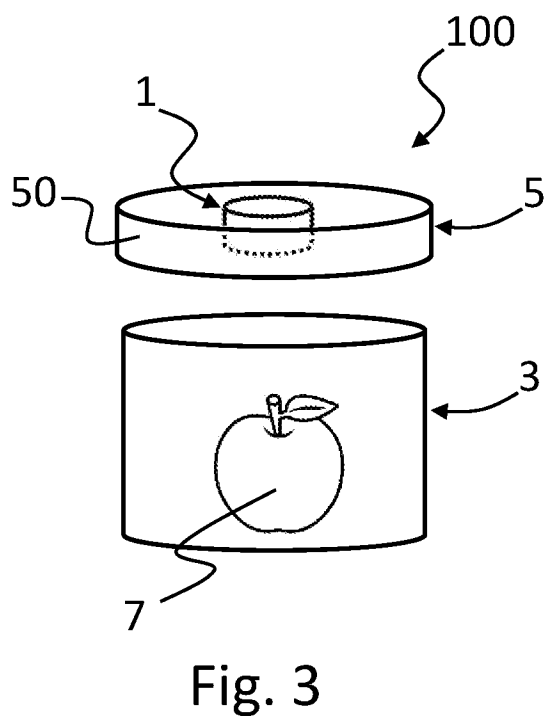
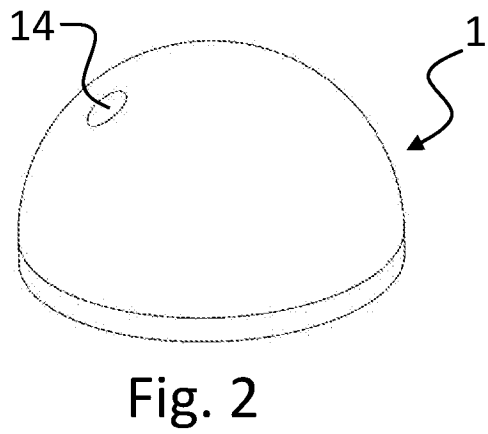
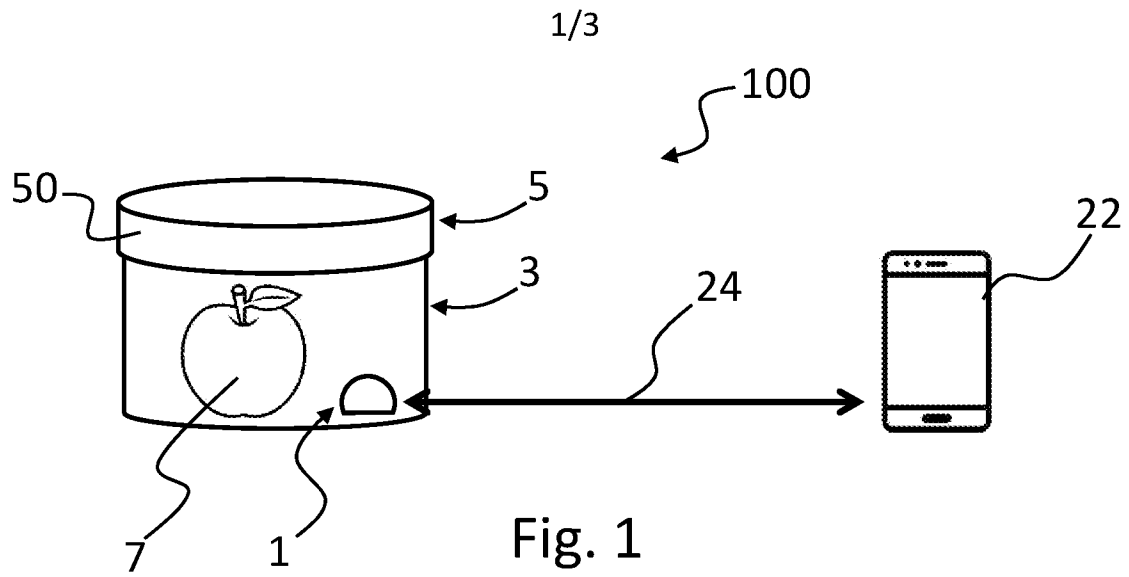
second classification algorithm being adapted to generate, on the basis at least of the variation over time of said signal indicative of the gas concentration, said signal indicative of the degree of freshness of said food (7) present in said container (3).

2. Device (1) according to the preceding claim, wherein, in at least said detection configuration, said device (1) is completely insertable inside said container (3).
3. Device (1), according to one or more of the preceding claims, characterised in that it is integrated in said closure (5) of said container (3) so that, when said closure (5) closes said container (3), said opening (14) faces said internal volume of said container (3).
4. Device (1), according to one or more of the preceding claims, comprising actuation means (18) configured to move said closing element (16) between an open position and a hermetically closed position of said opening (14).
5. Device (1) according to one or more of the preceding claims, wherein said processing and control unit (20) is configured to transmit to an external electronic device (22), through a data communication interface (24), said signal indicative of the gas concentration, and/or said signal indicative of the type of food (7) present in said container (3) and/or said signal indicative of the degree of freshness of said food (7) present in said container (3).
6. Device (1), according to one or more of the preceding claims, in which said gas detection device (9) comprises a volatile organic compound sensor of the type made of silicon with highly sensitive metal oxide (MOx) chemiresistor.
7. Device (1) according to one or more of the preceding claims, comprising a temperature sensor (26) configured to generate a signal indicative of the temperature of the gases present inside said cavity (12), said processing and control unit (20) being in data communication with said temperature sensor (26) and being configured to generate, on the basis of said signal indicative of the gas concentration and said signal indicative of the temperature of the gases, a signal indicative of the type of food (7) present in said container (3) and a signal indicative of the degree of freshness of said food (7) present in said container (3).
8. Device (1) according to one or more of the preceding claims, comprising a humidity sensor (28) configured to generate a signal indicative of the humidity of the gases present inside said cavity (12), said processing and control unit (20) being in data communication with said humidity sensor (28) and being configured to generate, on the basis of said signal indicative of the gas concentration and said signal indicative of the humidity of the gases, a signal indicative of the type of food (7) present in said container (3) and a

signal indicative of the degree of freshness of said food (7) present in said container (3).

9. Device (1), according to one or more of the preceding claims, wherein said signal indicative of the type of food (7) present in said container (3) comprises a signal indicative of a category, to which said food (7) belongs, identified among a plurality of predefined categories, and preferably also a signal indicative of a sub-category, to which said food (7) belongs, identified among a plurality of predefined sub-categories belonging to a respective category.
10. Device (1) according to one or more of the preceding claims, comprising a selector (8) operable by a user to select the category to which the food (7) present in said container (3) belongs among a plurality of preset categories of belonging, said processing and control unit (20) being configured to receive as input an initial signal indicative of the category to which the food (7) belongs selected by means of said selector (8) among said plurality of preset categories, said first classification algorithm and said second classification algorithm being configured to generate respectively said signal indicative of the type of food (7) and said signal indicative of the degree of freshness of said food (7), on the basis of said initial signal indicative of the category to which the food (7) present in said container (3) belongs, among said plurality of preset categories.
11. Device (1), according to one or more of the preceding claims, wherein said first and said second classification algorithm comprise artificial intelligence algorithms of the machine learning type.
12. Device (1), according to one or more of the preceding claims, wherein said first and second classification algorithm are configured to process comparisons on the basis of databases in which comparison data related to the gas emissions of known and predefined food are stored and provided.
13. Food preservation system (100) comprising at least one device (1) according to one or more of claims 1 to 12, and further comprising at least one food container (3) which can be closed by means of a closure (5).
14. Food preservation system (100) according to the preceding claim, wherein said closure (5) comprises a lid (50) comprising a seat (53) configured to accommodate said device (1).
15. Procedure for the recognition of a food and for the determination of the degree of freshness thereof, comprising the steps of:
- arranging a device (1) according to one or more of claims 1 to 12;
 - arranging a container (3) containing a food (7);

- setting said device (1) in said setting configuration for a predetermined period of time comprised between 30 seconds and 1 hour;
 - inserting, at least partially, said device (1) inside said container (3) containing said food (7);
- 5
- setting said device (1) in said detection configuration;
 - generating a signal indicative of the type of food (7) present in said container (3);
 - generating a signal indicative of the degree of freshness of said food (7) present in said container (3).



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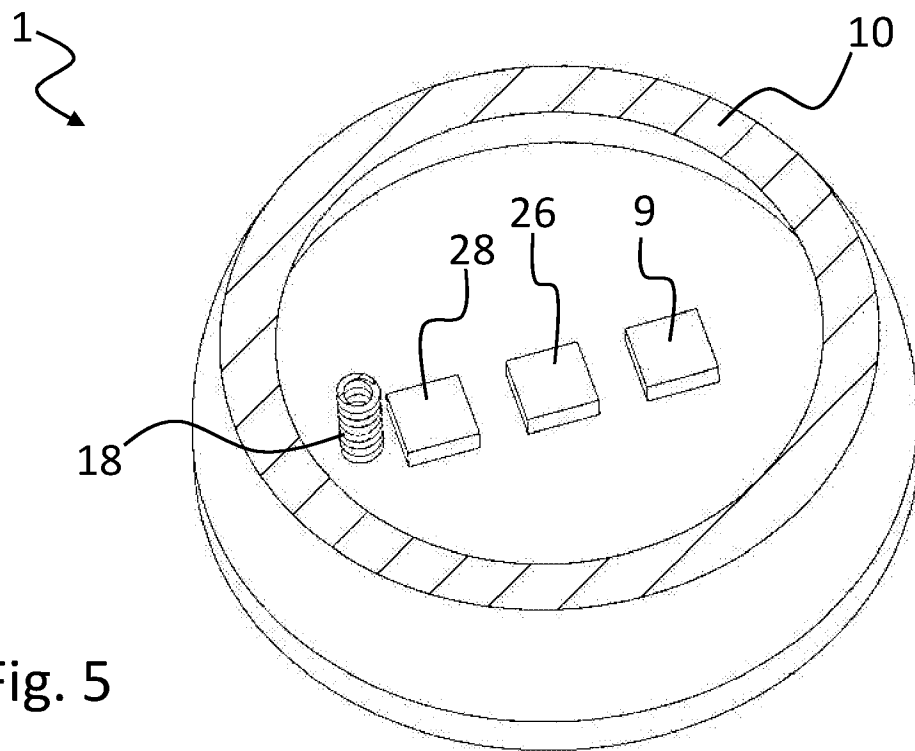


Fig. 5

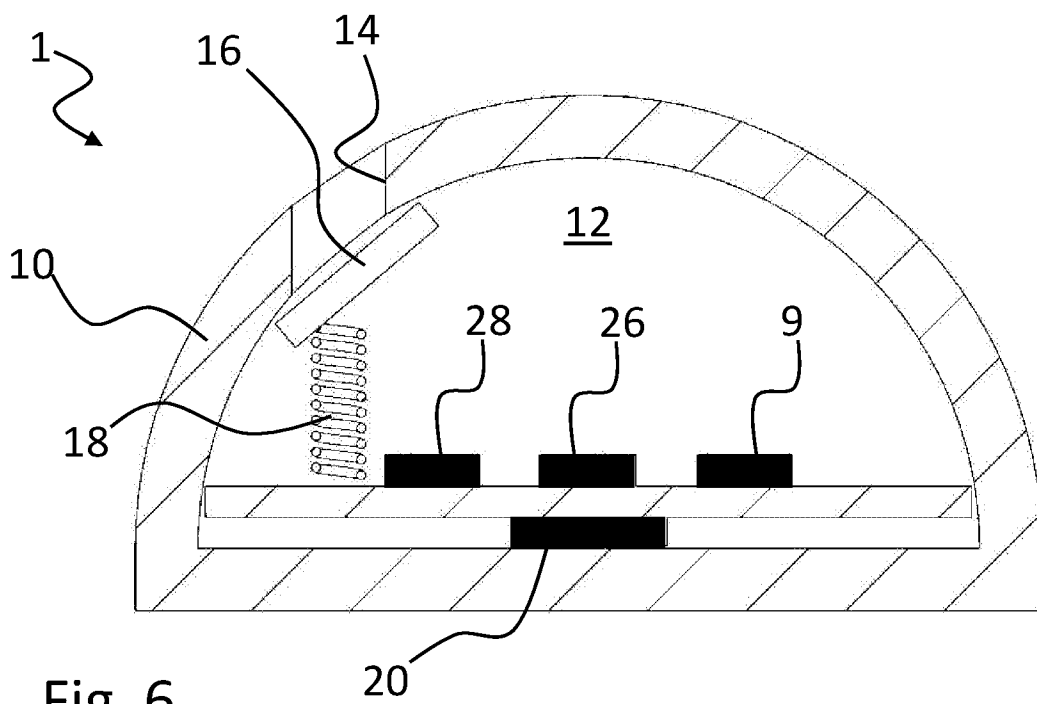


Fig. 6

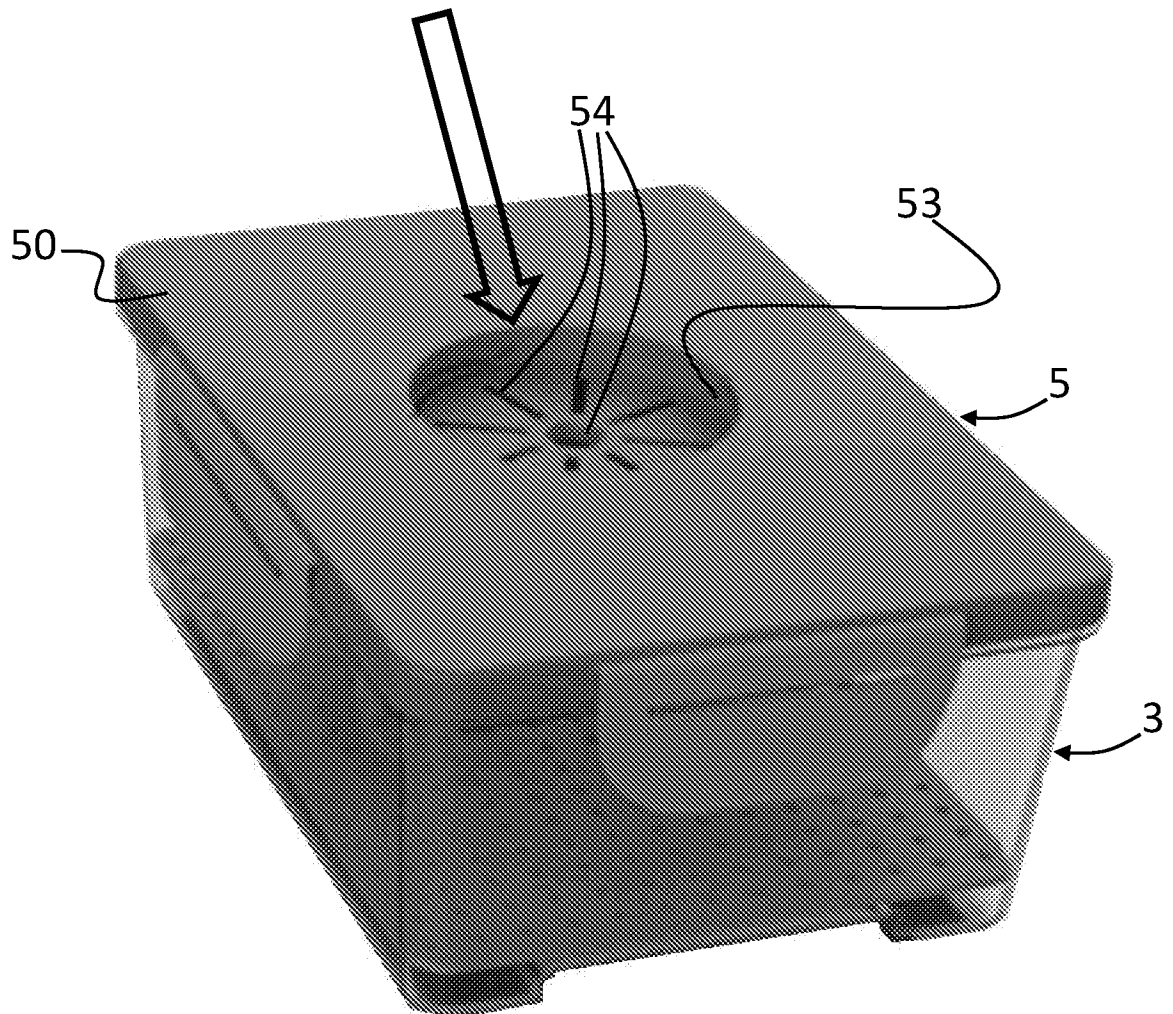
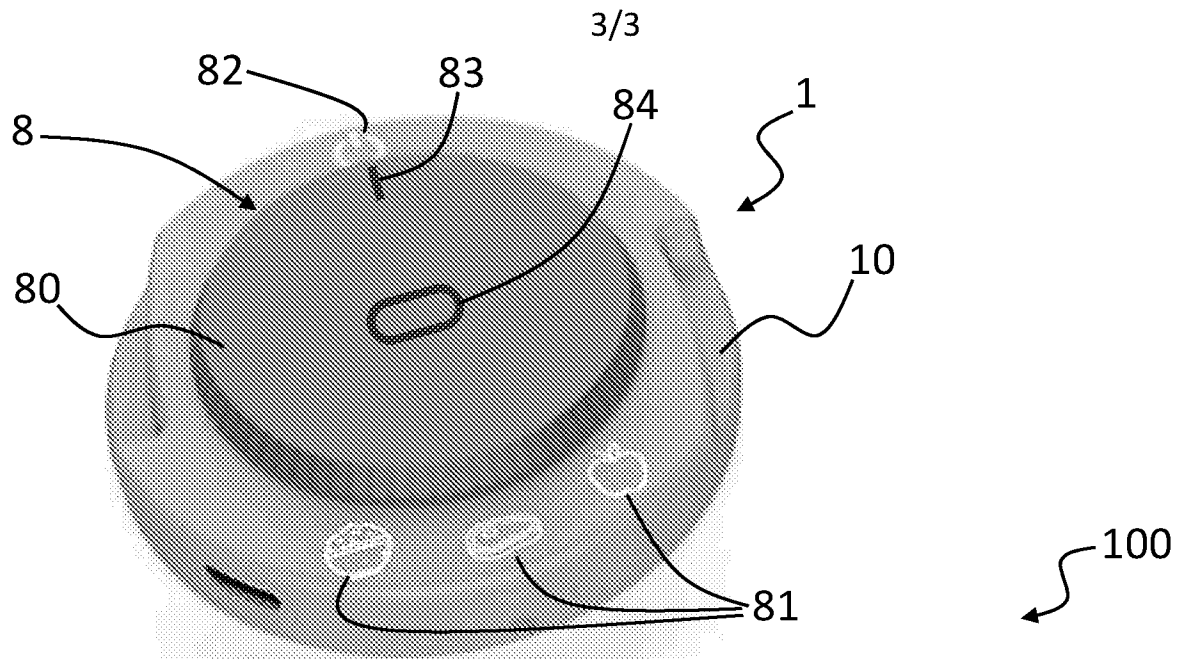


Fig. 7

INTERNATIONAL SEARCH REPORT

International application No

PCT/IB2024/052769

A. CLASSIFICATION OF SUBJECT MATTER INV. G01N33/00 ADD.		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED		
Minimum documentation searched (classification system followed by classification symbols) G01N F25D		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal, WPI Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2021/055275 A1 (PARK JANGPYO [KR] ET AL) 25 February 2021 (2021-02-25) paragraph [0005] - paragraph [0030]; figures 1-11 paragraph [0067] - paragraph [0122] paragraph [0152] - paragraph [0168] -----	1-15
X	US 2020/333310 A1 (LEE KYOUNGHOON [KR] ET AL) 22 October 2020 (2020-10-22) paragraph [0044] - paragraph [0087]; figures 1-6 -----	1-15
A	US 2021/025861 A1 (DAS SIDDHARTHA [US] ET AL) 28 January 2021 (2021-01-28) paragraph [0038] - paragraph [0047]; figure 1 -----	1-15
☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search 14 May 2024		Date of mailing of the international search report 27/05/2024
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Stavroulakis, E

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/IB2024/052769

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
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US 2021025861 A1	28-01-2021	NONE	